OpenDRIVE® 1.8.1, Static Road Network Description, 2024-11-21

13.12 Bridges

Bridges are modeled as objects in ASAM OpenDRIVE. The road with the bridge object leads over a bridge. By definition, bridges are valid for the complete cross section of a road. Bridges are described by a starting point, a length, and a type, such as concrete, steel, wood, or brick.

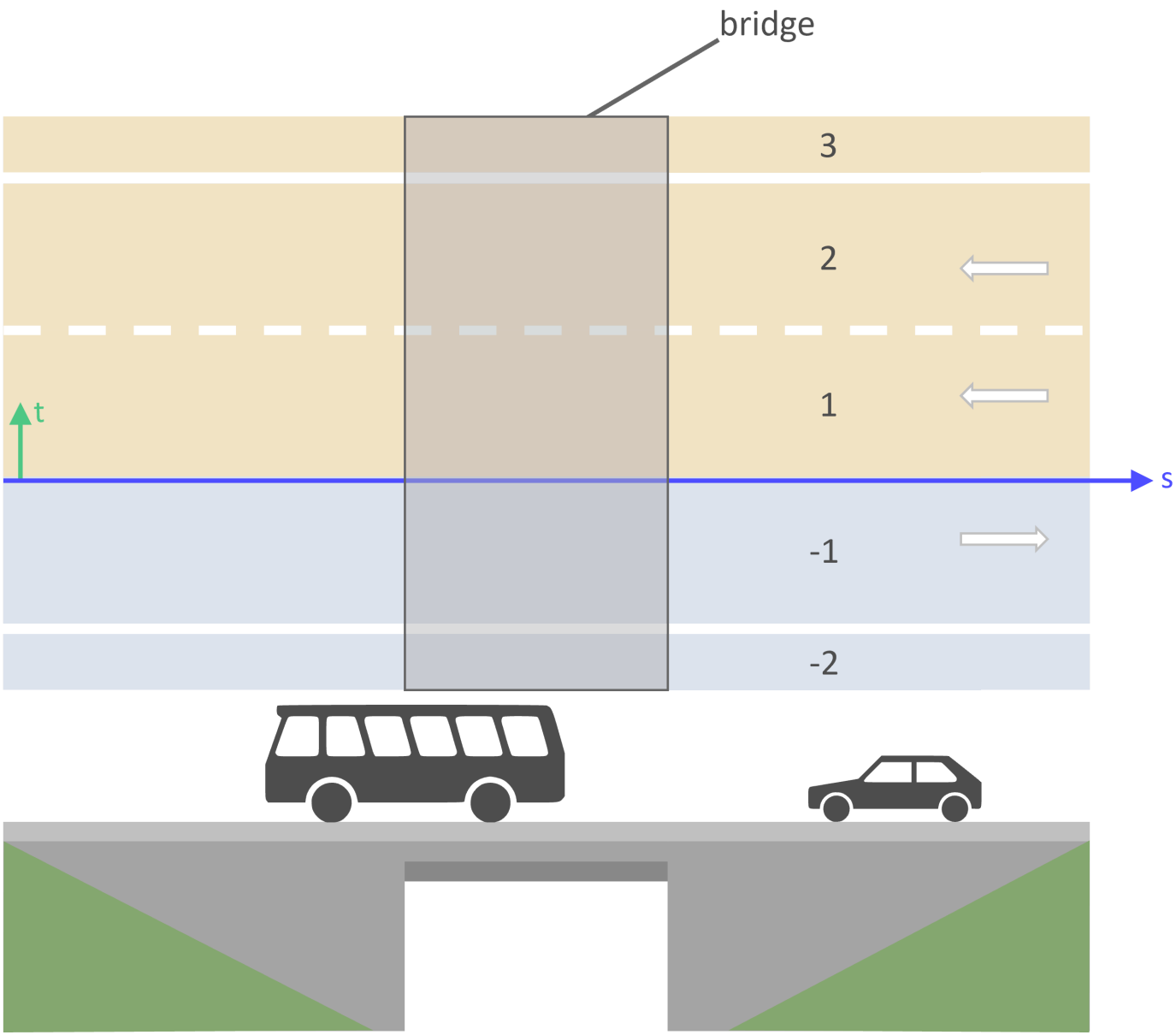

Figure 122. Bridge

Figure 122 shows a bridge that is valid for the whole cross section of the road and that defines the part that is the bridge.

Elements in UML model

`<bridge>` element

In ASAM OpenDRIVE, bridges are represented by the `<bridge>` element within the `<objects>` element.

```
UML class: t_road_objects_bridge
XML tag:   <bridge> (Multiplicity: 0..*)
```

Bridges are modeled as objects in ASAM OpenDRIVE. The road with the bridge object leads over a bridge. Bridges are valid for a road's complete cross section unless a lane validity record with further restrictions is provided as child element.

Table 104. Attributes of the <bridge> element

Name	Type	Use	Unit	Description
`id`	string	required		Unique ID within database
`length`	t_grEqZero	required	m	Length of the bridge (in s-direction)
`name`	string	optional		Name of the bridge. May be chosen freely.
`s`	t_grEqZero	required	m	s-coordinate
`type`	e_bridgeType	required		Type of bridge

`<validity>` element

In ASAM OpenDRIVE, lane validity is represented by the `<validity>` element within the `<object>` element.

```
UML class: t_road_objects_object_laneValidity
XML tag:   <validity> (Multiplicity: 0..*)
```

Lane validities restrict signals and objects to specific lanes.

Table 105. Attributes of the <validity> element

Name	Type	Use	Description
`fromLane`	integer	required	Minimum ID of the lanes for which the object is valid
`toLane`	integer	required	Maximum ID of the lanes for which the object is valid

XML example

```
<objects>
    <bridge s="50.0 "
            length="100.0"
            name="ExampleBridge"
            id="1"
            type="concrete"/>
</objects>
```

Rules

The following rules apply to bridge elements:

- asam.net:xodr:1.7.0:road.object.bridges.define_type: Bridges may be restricted to certain lanes, using the `<laneValidity>` element.
- asam.net:xodr:1.7.0:road.object.bridges.type_definition: The @type of the tunnel shall be specified.
- asam.net:xodr:1.7.0:road.object.bridges.from_lower_equal_to: The value of the @fromLane attribute shall be lower than or equal to the value of the @toLane attribute.

Related topics

- Section 10.2, "Properties for road sections and cross section"
- Section 13.6, "Lane validity for objects"